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Optimal scheduling of an integrated energy system considering carbon trading mechanism and multi-energy supply uncertainties in a real-time price environment

Abstract

To further constrain the carbon emission of the integrated energy system (IES) and reduce the comprehensive cost, an optimal scheduling model to balance economy and low carbon in a real-time price (RTP) environment was proposed. In this model, a low carbon operation strategy considering a reward and punishment ladder-type carbon tax, uncertain renewable power generation and uncertain load for the IES was proposed. First, a ladder-type carbon trading mechanism was introduced to participate in the carbon market to control carbon emissions, a RTP model with piecewise linear functions between this constructed IES and the main grid was considered, and then, the operation strategy of adjustable thermoelectric and electric cooling ratio was proposed to achieve flexibility and economy of IES operation. Next, the objective function was constructed using the deviation preference optimization strategy to reflect the different optimization preferences of the decision-makers, and the original model was converted into a mixed-integer linear programming by a series of mathematical processing and transformations. Finally, the simulation shows that the IES operation effectively balances economy and reliability by setting an appropriate confidence level for the spinning reserve, a ladder-type carbon tax and EV charging in IES has a considerable environmental and low-carbon advantages.

Keywords: Integrated energy system; Real-time price; Piecewise linear function; Ladder-type carbon trading mechanism; Adjustable thermoelectric and electric cooling ratio

1. Introduction

1.1. Background

Since the industrial revolution, global climate warming caused by greenhouse gas (GHG) emissions has become one of the most serious challenges to human society. There is a global consensus on action to address climate change, transitioning to a low-carbon socio-economic structure, most countries have signed the Paris Agreement to work together to limit global temperature rise to 2 °C by the end of the 21st

Nomenclature

Abbreviations

IES	Integrated energy system	AC	Absorption chiller
RES	Renewable energy sources	WT	Wind turbine
REPG	Renewable energy power generation	ELZ	Electrolyzer
CCHP	Combined cooling heating and power	GB	Gas boiler
DER	Distributed energy resources	HST	Hydrogen storage tank
CDER	Carbon dioxide emissions reduction	DR	Demand response
EV	Electric vehicle	FCP	Fuel cell
DR	Demand response	RT	Real-time price/pricing
PMSC	Power market scheduling center	SET	Sequence operation theory
DCE	Daily carbon emissions	CCP	Chance-constrained programming
DCC	Daily comprehensive cost	CGT	Conventional gas turbine
MILP	Mixed-integer linear programming	EE	Electrolyzer
HESS	Hybrid energy storage system	GB	Gas boiler
MCS	Monte carlo simulation	P2G	Power-to-gas
HFC	Hydrogen fuel-cell	PGST	Power-to-gas-to-storage tank
LTCT	Ladder-type carbon tax	PDR	Price-based demand response
LTCTM	Ladder-type carbon trading mechanism	SOC	State of charge
		DST	Discretized step
		DEF	transformation
		PDF	Dumped energy proportion
		PWLF	Probability density function
		SOC	Piecewise linear function
		EVG	State of charge
		MR	Electric vehicle group
			Methane reactor

century and strive to limit it to 1.5 °C. Many countries have set a clear timeline to achieve the goal of reducing GHG, for example, China has set an important strategic goal of reaching peak carbon by 2030 and carbon neutrality by 2060, Japan has also submitted Intended Nationally Determined Contribution to reduce GHG emissions by about 26% compared to 2013 levels by 2030, and reduce GHG emissions by at least 80% by 2050 (Chaube et al., 2021). Moreover, energy consumption will keep growing to face the critical phase of industrialization and urbanization. In this era of global low-carbon development, an integrated energy system (IES) is full of prospects for reducing carbon emissions (Z. Zhang et al., 2020). The IES couples a variety of energy sources for joint supply, formed by the organic coordination and optimization of various energy generation, transmission and distribution, conversion, storage and trading, which can

meet the terminal multi-energy and load demand, improves energy utilization, and promotes the consumption of renewable energy in the process of planning, construction and operation. Optimal scheduling of IES is a prerequisite for energy generation, utilization and balancing energy supply and demand. In the context of the energy crisis and environmental pollution, all countries around the world are desiring to improve energy utilization efficiency and optimize the energy flexibility while ensuring stable system operation (Ali et al., 2022; Gungor & Dincer, 2021). Nowadays, IES has become a hot spot and has been rapidly promoted. How to further constrain carbon emissions, cope with the uncertainty of renewable energy power generation (REPG), and play an active role in carbon tax regulation in the real-time price (RTP) environment has become the focus of our IES research.

1.2. Literature review

Most of the current literature on IES scheduling has focused on economic scheduling (Das & Hasan, 2021; Jiang et al., 2021; D. Yang et al., 2021), ignoring the great potential of IES to achieve carbon reductions, and the literature considering scheduling uncertainty, low-carbon operation, and system operational flexibility in a RTP environment simultaneously is relatively limited.

The existing literature rarely considers IES scheduling in a RTP environment. RTP has been widely used in demand-side management of power systems as an effective way of PBDR. Samadi et al. (2010) were the first to construct the social welfare model, which was the basis to apply the dual method to RTP research. To better characterize the generality of utility and cost functions, the method of piecewise linear function (PWL) was proposed in RTP studies (Xu et al., 2019). A RTP model for a smart power system considering joint welfare maximization for a microgrid system based on clean energy generation combined with a supplier based on fossil energy generation was constructed (Qu et al., 2022). An RTP strategy for the smart grid was designed which integrates multi-energy generation on the supply side, considering the interest of both supply and demand side, a bilevel stochastic model for real-time demand response (DR) was developed in the framework of markov decision process by a reinforcement learning approach (L. Zhang et al., 2022). In the recent literature on scheduling optimization, a bilevel optimal scheduling model considering the RTP mechanism was proposed to facilitate the economic operation of the battery-swapping stations involved in regulating the isolated microgrid (Y. Li et al., 2022).

The literature on uncertainty modeling for IES scheduling mainly includes stochastic optimization, robust optimization (Liu et al., 2021), and fuzzy programming (Liu et al., 2022). A tool of linear robust optimization combining uncertainty in power market prices and wind power generation was developed, and an uncertainty budget model that considers positive and negative deviations in power market prices was also proposed to improve the robustness against the errors in predicting uncertainty sources (Najafi et al., 2022). A bilevel robust optimization model for DR and thermal comfort was proposed for the capacity planning and operation of IES, taking into account the uncertainty of multi-energy load, renewable energy forecasts and the dissatisfaction of residents' participation in DR when reducing carbon emissions (X. Yang et al., 2021). A distributed robust optimization (DRO) scheduling model for a coupled electric-gas IES considering wind power uncertainty and P2G technology was proposed, and the proposed DRO model describes the uncertainty through a fuzzy set constructed based on the confidence bands of its probability density function, aiming to minimize the expected distribution of scheduling costs in the worst case (Y. Zhang et al., 2020). A combined framework of multi-objective optimization and robustness analysis was proposed for the optimal design of integrated building energy systems, addressing the trade-off between uncertainty and optimal robustness. (M. Wang et al., 2020). An IES stochastic optimization model that takes into account the dynamic characteristics of the network and psychological preferences was proposed (Zhong et al., 2020). An integrated stochastic planning-information gap decision theory (IGDT) approach was presented to handle multiple uncertainties in IES operations and planning optimization (Sun et al., 2022).

The low-carbon operation of IES has also attracted widespread attention. Scheduling optimization considering a reward and punishment ladder-type carbon tax (LTCT) was studied in a complicated community-integrated energy system (Zhu et al., 2023). Carbon neutral cost and DR were introduced into the scheduling strategy of IES through the carbon emission calculation method and carbon neutral cost model (Lv et al., 2022). Based on system-of-systems perspective, a model of carbon quota allocation and trading considering carbon balance and multi-system carbon sharing was developed (Huang et al., 2022). A LTCT has become the focus of low-carbon research, for example, An IES that calculates the carbon

trading cost and considers carbon capture technologies through a reward and punishment ladder-type carbon trading mechanism (LTCTM) with rewards and penalties was created (R. Wang et al., 2022).

In the scheduling model of IES, some studies consider operation flexibility. For example, by considering fast-responding flexible loads, power-to-gas (P2G) technologies, electric vehicles (EVs), and gas storage systems, a bilevel programming framework for resilience enhancement of electricity-gas-heat networks integrated with energy centers was proposed (Javadi et al., 2022). An architecture for an IES for coal mines was proposed that integrates these forms of related energy sources with flexible loads to improve system flexibility (Hu et al., 2022). System flexibility was improved by determining the optimal sizing of Power-to-X facilities, thereby optimizing IES operations (Zhan et al., 2022). The coordination of flexible resources between different energy carriers to accommodate different levels of wind power enables the integration of gas, heat, and power systems and multiple options for meeting customer energy needs (Xi et al., 2021). Most of the existing literature assumes a constant thermoelectric ratio, which leads to a mismatch between the output of Combined cooling heating and power (CCHP) and the actual energy demand, therefore, the adjustable thermoelectric and electric cooling ratio of CCHP was proposed to improve the energy supply efficiency and optimize the economic operation of IES (J. Li et al., 2019).

The above literature has conducted relevant studies in terms of RTP, uncertainties, low-carbon operation, and flexibility, respectively. Promoting renewable energy consumption is an effective way to save energy and reduce emissions. However, the peak period of wind power output and grid electricity consumption do not coincide, and electricity cannot be stored on a large scale yet, which leads to wind abandonment. RTP is the most ideal pricing method under a smart grid, an effective demand-side management method, and an effective means to regulate customers' electricity consumption behavior through the market mechanism. Through the RTP mechanism, it guides the behaviors of customers' consumption, achieves the purpose of peak-shaving and valley-filling of electric energy, and realizes the effective use of electric power resources. The popularization of renewable energy generation has a positive role in promoting the development of hydrogen energy. The development of hydrogen energy not only reduces the waste of electric energy but also minimizes the fluctuation of the power grid. Meanwhile, as the cost becomes lower and lower, the large-scale application of hydrogen energy storage will replace

traditional electric energy storage equipment. For the phenomenon of renewable energy abandonment, excess power into natural gas or hydrogen is converted by P2G technology, which greatly increases the consumption ability of wind power (X. Zhang et al., 2019). The P2G process contains two processes: electrical power to hydrogen by electrolysis and hydrogen to methane either catalytically or biologically through reaction with CO₂ (Ahern et al., 2015). Compared to the efficiency of electricity to natural gas, the efficiency of electricity to hydrogen will be much higher while the combustion is more efficient and does not produce any pollutant emissions.

Since the LTCT can be viewed as a cost PWLF, as well as various utilities can be also seen as a utility PWLF, a scheduling optimization study considering a PWLF is introduced in a RTP environment based on the literature (Xu et al., 2019). Based on the above, we integrate low-carbon operation, the flexibility of refined P2G operation, and CCHP with adjustable thermoelectric and electric cooling ratios on IES scheduling, consider the impact of multiple uncertainties and carbon tax on IES in the RTP environment, and propose a LTCTM for the carbon trading market, as shown in Table 1 below.

Table 1 Summary of recent research on IES scheduling optimization

Reference	Uncertainty			RTP	Low-carbon	Flexibility	Spinning reserve	EV
	WT	PV	Load					
(Xu et al., 2019)	×	×	×	✓	×	×	×	×
(Y. Li et al., 2022)	✓	✓	✓	✓	×	×	✓	✓
(Qu et al., 2022)	×	×	×	✓	✓	×	×	×
(L. Zhang et al., 2022)	✓	✓	✓	✓	×	✓	×	×
(Najafi et al., 2022)	✓	×	×	×	×	×	×	×
(X. Yang et al., 2021)	✓	✓	✓	×	×	×	×	×
(Y. Zhang et al., 2020)	✓	×	×	×	×	✓	×	×
(M. Wang et al., 2020)	✓	✓	✓	×	✓	×	×	×
(Zhong et al., 2020)	✓	×	✓	×	×	✓	×	×
(Sun et al., 2022)	×	✓	✓	×	×	×	×	×
(Zhu et al., 2023)	✓	✓	✓	×	✓	✓	✓	✓
(Lv et al., 2022)	✓	✓	✓	×	✓	✓	×	×
(Huang et al., 2022)	×	×	×	×	✓	×	×	×
(R. Wang et al., 2022)	✓	×	×	×	✓	×	×	×
(Javadi et al., 2022)	×	×	×	×	×	✓	×	✓
(Hu et al., 2022)	✓	✓	✓	×	✓	✓	×	×

(Zhan et al., 2022)	✓	×	✓	×	×	✓	×	×
(Xi et al., 2021)	✓	×	✓	×	×	×	×	×
(J. Li et al., 2019)	×	✓	✓	✓	×	✓	×	×
This work	✓	✓	✓	✓	✓	✓	✓	✓

1.3. Works and contributions

Our main works and contributions were summarized below:

- 1) RTP in IES scheduling was introduced, where users in IES and the external grid supplier exchange electricity dynamically in a RTP environment with the development and popularity of smart grid.
- 2) A reward and punishment ladder-type carbon trading mechanism (LTCTM) was introduced to reflect and regulate the current carbon emission market quota system.
- 3) Piecewise linear functions (PWL) were introduced to accurately portray the cost of renewable energy power generation and further constrain the carbon emissions by a reward and punishment ladder-type carbon tax, moreover, the smoothing method of PWL was theoretically proven to be effective and used to solve the model.
- 4) An IES low-carbon operation strategy considering multiple uncertainties of renewable energy generation, EVs and load was proposed.
- 5) The operation strategy of adjustable thermoelectric and electric cooling ratios was introduced to reflect the flexibility of IES operation in CCHP and hydrogen fuel-cell (HFC) subsystems.

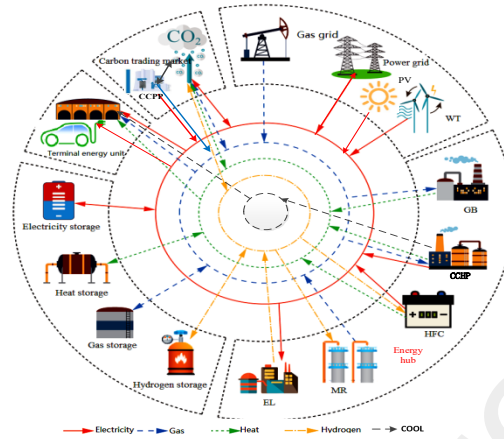


Figure 1 Operation diagram of the IES.

2. Mathematical model

The IES low-carbon economy scheduling framework mainly contains five parts, shown in Figure 1: energy supply subsystem, coupling device subsystem, energy storage device subsystem, end-use energy subsystem, and carbon trading market subsystem. The distributed power supply provides power for IES; the electrolyzer (EL) converts power into hydrogen, which is further converted into natural gas through the Methane Reactor (MR), and can also be directly supplied to HFC, improving the energy utilization; the gas boiler burns natural gas to meet the thermal load demand; CCHP burns natural gas to meet both power and thermal load demand; gas load demand is jointly supplied by the gas network or MR; in addition, the IES also contains electricity, gas, thermal and hydrogen energy storage device for storage, the carbon dioxide absorbed or emitted in the operation of each device is ultimately traded through the carbon market.

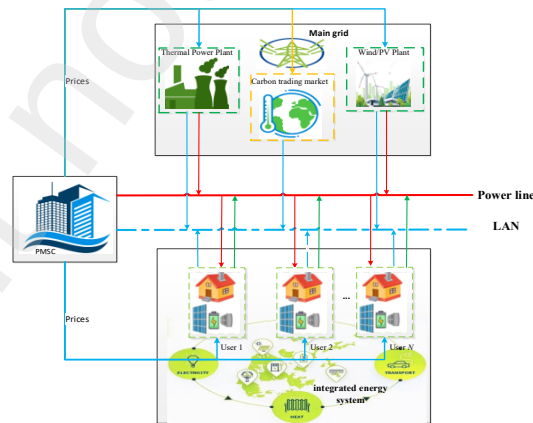


Figure 2 Smart grid with carbon trading market and multi-energy generation.

2.1. RTP based on the social welfare model

To get the RTP for electricity in IES, we introduce a social welfare model, which is a widely used model in the field of RTP that considers the maximization of both supply-side and demand-side benefits. Based on the social welfare maximization model in (Samadi et al., 2010), we can obtain RTP by the Lagrange multiplier and dual method . The original social welfare model is shown below:

$$\begin{aligned} \max_{L_i^{\max} \leq L_i \leq L_i^{\max}} \sum_{t \in T} \sum_{i \in N} U(x_i^t, \omega_i^t) - C_i(L_i) \\ \text{s.t.} \quad \sum_{i \in N} x_i^t \leq L_t, \quad \forall t \in T \end{aligned} \quad (1)$$

where $U(x_i^t, \omega_i^t)$ and $C_i(L_i)$ denote the utility for user i and the cost for supplier provided electricity L_i at time slot t , ω_i^t is the elastic coefficient for user i at time slot t . Different users are distinguished at different time slots by different elastic coefficients. In general, the utility function is characterized by some properties from (Samadi et al., 2010; Y. Yang et al., 2023), which can be expressed as the quadratic function (Samadi et al., 2010; Deng et al., 2015), the logarithmic function (Song & Qu, 2014), the hyperbolic function (Yusta et al., 2007) and the PWLF (Xu et al., 2019), etc. Since Problem (1) is a convex optimization, it can be solved by a dual method, and the Lagrange function of Problem (1) is obtained by introducing a Lagrange multiplier to represent the electricity price, the iterative process of subgradient projection and the algorithmic flow of RTP are referred to the literature (Samadi et al., 2010). The interaction process between this IES and the external power grid is in a RTP environment, and the framework of the interaction welfare algorithm is shown in Figure 2.

2.2. Piecewise linear function

To characterize the utility and cost functions in a rational and refined manner, we propose a piecewise linear function (PWLF). It consists of n components, each of which is a linear function. Utility is used to measure personal requirements and desires on the basis of consumption or leisure activities, which is determined by the intensity of consumer desire. According to mathematical theory, a quadratic or smooth function cannot precisely characterize utility. Meanwhile, the PWLF can approximate any function well,

so taking the piecewise linear or smooth function as a utility function is an effective approach (Xu et al., 2019; Gao, 2022). Therefore, a PWLF is suitable to represent and characterize both utility and cost functions. Although both the utility and cost functions can be portrayed by different quadratic functions (Samadi et al., 2010), the utility preferences of different users are different, and the utility preferences of the same user at different times are also different. Meanwhile, with the continuous increase of renewable energy penetration rate, the quadratic cost function to generate electricity solely by traditional thermal plants is no longer applicable, therefore, the cost or utility functions inscribed by quadratic functions alone are not refined enough to accurately reflect the actual levels. To characterize the quadratic function perfectly, we propose a PWLF, which contains n segments, formulated as follows:

$$f(x) = \begin{cases} f_0(x) = \omega_1 x, & x \in [-a_1, a_1] \\ f_i(x) = \omega_{i+1} x + \sum_{u=1}^i ((\omega_u - \omega_{u+1}) a_u), & x \in (a_i, a_{i+1}] \cup [-a_{i+1}, -a_i) \quad \forall i = 1, \dots, n-1 \end{cases} \quad (2)$$

where $a_1 < a_2 < \dots < a_n$, $f_i(x)$ is a continuously differentiable function, $f(x)$ is continuous but not smooth at a_i . When characterizing the utility function, $\Delta\omega_i = \omega_{i+1} - \omega_i < 0$, $x > 0$, x denotes the electricity consumption of users. When characterizing the cost function, $\Delta\omega_i = \omega_{i+1} - \omega_i > 0$, if portrayed as the generation cost, $x > 0$, x denotes the electricity generation of suppliers, if portrayed as the carbon tax, x denotes the difference between carbon emissions and carbon quotas, x can be negative, $f(x) = -f(-x)$.

To ensure that the welfare maximization model is a solvable convex optimization problem, we need to smoothing the PWLF. The PWLF is non-smooth, so a smoothing method is introduced to construct a smoothing function to approximate the PWLF. Therefore, the global smoothing function $\hat{f}_\varepsilon(x)$ to approximate $f(x)$ at the interval partition point $a_i (i = 1, \dots, n)$ is below:

$$\hat{f}_{\varepsilon_i}(x) = \frac{f_{i+1}(x) + f_{i+2}(x)}{2} + \frac{(x - a_{i+1})(f_{i+2}(x) - f_{i+1}(x))}{2((x - a_{i+1})^2 + \varepsilon)^{\frac{1}{2}}}, \quad (3)$$

$$x \in (a_{i-1}, a_i] \cup [-a_i, -a_{i-1}) \quad \forall i = 1, \dots, n-1$$

when the smooth coefficient ε is small enough, $\lim_{\varepsilon \rightarrow 0^+} \hat{f}_\varepsilon(x) = f(x)$. The mathematical derivation of the error $\left| \hat{f}_{\varepsilon_i}(x) - f_i(x) \right|$ is shown in Appendix A.1. From the derivation, we draw the following conclusions:

$$\left| \hat{f}_{\varepsilon_i}(x) - f_i(x) \right| = \left| \hat{f}_{\varepsilon_i}(x) - f_i(x) \right| = \frac{1}{2} \left| \frac{(\omega_{i+1} - \omega_i)\varepsilon}{x - a_i} \right| \quad (4)$$

At the segment point $x = a_i$, $f_1(a_1) = \omega_1 a_1 = f_2(a_1)$, since $\frac{df_1(x)}{dx} = \omega_1 \neq \frac{df_2(x)}{dx} = \omega_2$, i.e., the PWLF $f(x)$ is continuously non-smooth at a_i . Similarly, the PWLF is continuously non-smooth at $a_2, \dots, a_n$. Therefore, $f(x)$ is not smooth at $x = a_i, i = 1, \dots, a_n$ and needs to be smoothed. Based on the above theory, the LTCT is considered as a piecewise cost function, the smoothing method can therefore be used in the LTCT.

2.3. A reward and punishment ladder-type carbon trading mechanism

Carbon trading is a mechanism to control carbon emissions by establishing legal carbon credits and allowing producers to trade them in the carbon market. In carbon trading, power producers use energy-saving technologies to reduce carbon emissions, and then sell the excess carbon quotas as a commodity for a profit. When carbon emissions are less than carbon quotas, the carbon tax is negative and has an incentive effect; when carbon emissions exceed carbon quotas, the carbon tax is positive and has a punitive mechanism. The regulator first allocates carbon emission quotas to each energy supplier, and the producer produces and emits them by their quotas. If the actual carbon emissions are lower than the allocated quotas, the remaining quotas are traded in the carbon trading market, and if not, carbon credits need to be purchased. The incentives and punishment LTCTM model mainly includes the carbon emission quota model, the actual carbon emission model, and the reward and punishment ladder-type carbon emission trading model. Ladder carbon trading divides carbon emission rights into several segments, and the more carbon quotas required for the interval, the higher the carbon trading price. This trading mechanism can promote energy laddering utilization, increase the benefits of energy saving and carbon emission reduction, and contribution to the achievement of low-carbon goals.

To give full play to the regulation and guidance of the market economy and further exploit the market potential of carbon emission reduction, we propose the concept of a LTCTM with incentives and penalties composed of PWLFs and give the theoretical derivation of the smoothing processing.

A. The carbon emission quota model

In our IES model, there are three different types of carbon emission sources: purchasing power, GB, and CCHP, and it is considered that the purchased power from the grid comes from coal-fired units.

$$\begin{cases} E_{IES} = E_{e, \text{buy}} + E_{CCHP} + E_{GB} \\ E_{e, \text{buy}} = \chi_e \sum_{t=1}^T P_{e, \text{buy}}(t) \\ E_{CCHP} = \chi_g \sum_{t=1}^T (P_{CCHP,e}(t) + P_{CCHP,h}(t) + P_{CCHP,c}(t)) \\ E_{GB} = \chi_g \sum_{t=1}^T P_{GB,h}(t) \end{cases} \quad (5)$$

where E_{IES} , $E_{e, \text{buy}}$, E_{CCHP} , and E_{GB} are the carbon emission quotas in IES, purchased electricity, CCHP, and GB; χ_e and χ_g are the carbon emission quotas per unit of electricity consumption of coal-fired units and per unit of natural gas consumption of natural gas-fired units; $P_{e, \text{buy}}(t)$ is the purchasing power at time slot t ; $P_{GB,h}(t)$ is the thermal energy output of GB at time slot t ; T is the scheduling period.

B. The actual carbon emission model

$$\begin{cases} E_{IES,a} = E_{e, \text{buy},a} + E_{total,a} - E_{MR,a} \\ E_{e, \text{buy},a} = \sum_{t=1}^T (a_1 + b_1 P_{e, \text{buy}}(t) + c_1 P_{e, \text{buy}}^2(t)) \\ E_{total,a} = \sum_{t=1}^T (a_2 + b_2 P_{total}(t) + c_2 P_{total}^2(t)) \\ P_{total}(t) = P_{CCHP,e}(t) + P_{CCHP,h}(t) + P_{GB,h}(t) \\ E_{MR,a} = \sum_{t=1}^T \varpi P_{MR,g}(t) \end{cases} \quad (6)$$

where $E_{IES,a}$, $E_{e, \text{buy},a}$ are the actual carbon emissions of IES and the purchasing power from the upper level; $E_{total,a}$ is the total actual carbon emissions of CCHP, GB, and MR; $E_{MR,a}$ is the actual amount of CO₂ absorbed by MR; $P_{total}(t)$ is the equivalent power output of CCHP, GB, and MR at time slot t ; a_1 , b_1 , c_1 and a_2 , b_2 , c_2 are the calculated parameters of carbon emissions of coal-fired and natural gas-fired energy supply equipment; ϖ is the CO₂ absorption of MR equipment during the conversion of hydrogen to natural gas.

C. The reward and punishment ladder-type carbon emission trading model

The actual amount of carbon credits traded for participation in the carbon trading market is equal to the actual carbon emissions minus the IES carbon quotas, expressed as follows:

$$E_{IES,t} = E_{IES,a} - E_{IES} \quad (7)$$

We adopt a reward and punishment LTCTM to further constrain carbon emissions in IES. The ladder pricing mechanism is divided into multiple purchase ranges, and the purchase price of the corresponding range increases as more carbon quotas are purchased. According to the derivation of Eq. (2), set $\Delta\omega_i = \alpha\lambda$, n identical interval partition points, the LTCT cost is below:

$$f_{CO_2} = \begin{cases} \lambda E_{IES,t} & , E_{IES,t} \in (-l, l] \\ \lambda(1+\alpha)(E_{IES,t} - l) + \lambda l & , E_{IES,t} \in (l, 2l] \cup [-2l, -l] \\ \lambda(1+(i-1)\alpha)(E_{IES,t} - (i-1)l) + \lambda \left((i-1)l + \alpha \sum_{u=1}^{i-2} l \right) & , \\ E_{IES,t} \in ((i-1)l, il] \cup [-il, -(i-1)l] , \forall i = 3, \dots, n \end{cases} \quad (8)$$

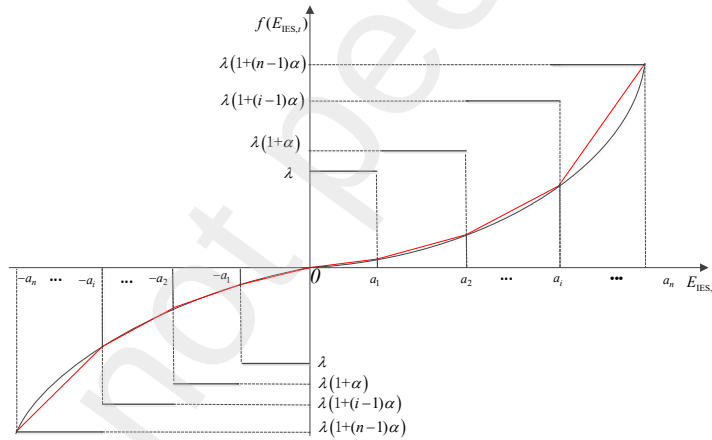


Figure 3 Piecewise linear function with slopes of the LTCT.

According to the derivation of Eq. (3), set $\Delta\omega_i = \alpha\lambda$, the smoothing function of the LTCT cost is below:

$$f_{\varepsilon_co_2}^{\wedge}(x) = \frac{\lambda}{2} \left[\frac{\left(2E_{IES,t} + (2i-1)\alpha E_{IES,t} - i^2\alpha l \right) + \alpha (E_{IES,t} - il)^2}{\left[(E_{IES,t} - il)^2 + \varepsilon \right]^{\frac{1}{2}}} \right], \quad (9)$$

$$E_{IES,t} \in ((i-1)l, il] \cup [-il, (1-i)l], \quad \forall i = 1, 2, \dots, n$$

where f_{CO_2} and $f_{\varepsilon_co_2}^{\wedge}$ is the real and the smoothing function of LTCT cost; λ is the carbon trading base price; α is the price growth rate; n is the number of pieces for reward or punishment ladder-type function; l is the length of the carbon emission interval.

2.4. Two-stage P2G operation model

Hydrogen energy is a zero-carbon emission and efficient clean energy source, which has advantages in cross-seasonal high-capacity energy storage and can be used to supply chemical industry products, hydrogen fuel cells or hydrogen vehicles. In this paper, hydrogen is produced by the electrolysis of water. First, the EL converts electrical energy into hydrogen energy, and part of the hydrogen is input into the MR to synthesize natural gas with CO_2 to supply natural gas load, CCHP, and GB, and part of the hydrogen energy is directly transferred to the HFC for converting into electricity and thermal, and the remaining part is stored in the hydrogen storage tank. Compared with the conversion to natural gas and then to GB or CCHP combustion, the direct conversion of hydrogen energy to electricity and heat through HFC reduces one energy conversion step and reduces the energy step loss, and the energy efficiency of hydrogen is much higher than that of conventional natural gas and no CO_2 is produced. The above energy conversion model is described as follows:

(1) EL modeling

$$\begin{cases} P_{EL,H_2}(t) = \eta_{EL} P_{e,EL}(t) \\ P_{e,EL}^{\min} \leq P_{e,EL}(t) \leq P_{e,EL}^{\max} \\ \Delta P_{e,EL}^{\min} \leq P_{e,EL}(t+1) - P_{e,EL}(t) \leq \Delta P_{e,EL}^{\max} \end{cases} \quad (10)$$

where $P_{e,EL}(t)$ is the electrical energy input to EL at the time slot t ; $P_{EL,H_2}(t)$ is the hydrogen energy output from EL at the time slot t ; η_{EL} is the energy conversion efficiency of EL; $P_{e,EL}^{\max}$ and $P_{e,EL}^{\min}$ are the power upper and lower limits of EL; $\Delta P_{e,EL}^{\max}$, $\Delta P_{e,EL}^{\min}$ are the upper and lower boundaries of climbing power of EL.

GB or MR modeling is similar to the above modeling.

(2) HFC modeling

$$\begin{cases} P_{\text{HFC},e}(t) = \eta_{\text{HFC}}^e P_{\text{H}_2,\text{HFC}}(t) \\ P_{\text{HFC},h}(t) = \eta_{\text{HFC}}^h P_{\text{H}_2,\text{HFC}}(t) \\ P_{\text{H}_2,\text{HFC}}^{\min} \leq P_{\text{H}_2,\text{HFC}}(t) \leq P_{\text{H}_2,\text{HFC}}^{\max} \\ \Delta P_{\text{H}_2,\text{HFC}}^{\min} \leq P_{\text{H}_2,\text{HFC}}(t+1) - P_{\text{H}_2,\text{HFC}}(t) \leq \Delta P_{\text{H}_2,\text{HFC}}^{\max} \\ \kappa_{\text{HFC}}^{\min} \leq P_{\text{HFC},h}(t) / P_{\text{HFC},e}(t) \leq \kappa_{\text{HFC}}^{\max} \end{cases} \quad (11)$$

where $P_{\text{H}_2,\text{HFC}}(t)$ is the hydrogen energy input to HFC at time slot t ; $P_{\text{HFC},e}$ and $P_{\text{HFC},h}$ are the electric and thermal energy output from HFC at time slot t ; η_{HFC}^e and η_{HFC}^h are the efficiency of converting HFC to electric and thermal energy; $P_{\text{H}_2,\text{HFC}}^{\max}$ and $P_{\text{H}_2,\text{HFC}}^{\min}$ are the upper and lower boundaries of hydrogen energy input to HFC; $\Delta P_{\text{H}_2,\text{HFC}}^{\max}$ and $\Delta P_{\text{H}_2,\text{HFC}}^{\min}$ are the upper and lower climbing power boundaries of HFC; $\kappa_{\text{HFC}}^{\max}$ and $\kappa_{\text{HFC}}^{\min}$ are the upper and lower boundaries of HFC thermoelectric ratio.

2.5. Adjustable thermoelectric and electric cooling ratio model

CCHP is a subsystem that generates electricity by burning natural gas, and at the same time, it can supply the heating and cooling load with the waste heat generated from the power generation process. The thermoelectric ratio is the ratio of heating load to power (converted into heat) in CCHP subsystem. Namely, it is the ratio of heat supplied to end-users to the output power (converted into heat) in CCHP subsystem. CCHP with adjustable thermoelectric and electric cooling ratio can adjust the ratio of electricity and heat output according to the real-time power and heat demand, thus optimizing the operating efficiency, and its mathematical model of CCHP operation is formulated below:

$$\begin{cases} P_{\text{CCHP},e}(t) = \eta_{\text{CCHP}}^e P_{\text{g,CCHP}}(t) \\ P_{\text{CCHP},h}(t) = \eta_{\text{CCHP}}^h P_{\text{g,CCHP}}(t) \\ P_{\text{CCHP},c}(t) = \eta_{\text{CCHP}}^c P_{\text{g,CCHP}}(t) \\ P_{\text{g,CCHP}}^{\min} \leq P_{\text{g,CCHP}}(t) \leq P_{\text{g,CCHP}}^{\max} \\ \Delta P_{\text{g,CCHP}}^{\min} \leq P_{\text{g,CCHP}}(t+1) - P_{\text{g,CCHP}}(t) \leq \Delta P_{\text{g,CCHP}}^{\max} \\ \mu_{\text{CCHP}}^{\min} \leq P_{\text{CCHP},h}(t) / P_{\text{CCHP},e}(t) \leq \mu_{\text{CCHP}}^{\max} \\ \nu_{\text{CCHP}}^{\min} \leq P_{\text{CCHP},c}(t) / P_{\text{CCHP},e}(t) \leq \nu_{\text{CCHP}}^{\max} \end{cases} \quad (12)$$

where $P_{\text{g,CCHP}}(t)$ is the natural gas power input to CCHP at time slot t ; $P_{\text{CCHP},e}(t)$, $P_{\text{CCHP},h}(t)$ and $P_{\text{CCHP},c}(t)$ are the electric, thermal and cooling energy output from CCHP at time slot t ; η_{CCHP}^e , η_{CCHP}^h and η_{CCHP}^c are the efficiency of converting CCHP to electric, thermal and cooling energy; $P_{\text{g,CCHP}}^{\max}$ and $P_{\text{g,CCHP}}^{\min}$ are the upper and lower boundaries of

natural gas power input to CCHP; $VP_{g,CCHP}^{\max}$ and $VP_{g,CCHP}^{\min}$ are the climbing upper and lower boundaries of CCHP power output; $\mu_{CCHP}^{\max}$ and $\mu_{CCHP}^{\min}$ are the upper and lower boundaries of CCHP thermoelectric ratio; $\nu_{CCHP}^{\max}$ and $\nu_{CCHP}^{\min}$ are the upper and lower boundaries of CCHP electric cooling ratio.

2.6. Uncertainties model

A. Probabilistic EV load model

The individual EV charging is random, however, it will show a specific distribution when a certain size of EVs is reached. The EV daily mileage D_{EV} satisfies a lognormal distribution (Qian et al., 2011), and the probability density function (PDF) of the starting time of charging t_s obeys piecewise normal distribution (Tian et al., 2010), shown in Figure 21 in Appendix B.2. The charging duration time t_c of each EV is $t_c = D_{EV} P_{pch} / 100 \delta_c P_c$ (Tian et al., 2010), where P_{pch} is the power consumption per hundred kilometers; δ_c denotes the EVs charging conversion efficiency; P_c denotes the charging power. The change in charging price is affected by their charging behavior, some EVOs can adjust the charging time by the electricity price under a time-of-use (TOU) pricing environment. The distribution function of the charging price of EVs is as follows:

$$c_p(t) = \begin{cases} p_{valley} & , t_{start} \leq t \leq t_{end} \\ p_{peak} & , \text{else} \end{cases} \quad (13)$$

where p_{valley} and p_{peak} denote the electricity price in the valley and peak hours, t_{start} and t_{end} denotes the moment when the valley time starts and ends. Assumptions include: (i) EVOs tend to charge vehicles at low electricity prices. (ii) No unexpected events occurring to affect EVOs. (iii) Charging power is uniformly distributed in the range of [3.5,7]; battery capacity is uniformly distributed in the range of [60,70]; (iv) PBDR scheme was designed in TOU prices environment. The proportion of EVOs who follow the PBDR is η ($0 < \eta < 1$) of the total EVOs during the valley hours; EVOs are divided into two parts, N (N_{PDR} plus N_{NPDR}) EVs: $N_{PDR} = N * \eta$, $N_{NPDR} = N * (1 - \eta)$, where N_{PDR} denotes the numbers of EVOs affected by charging prices and N_{NPDR} represents those not affected. (v) The starting time not affected by PBDR obeys a normal distribution, while affected by PBDR obeys a uniform distribution. (vi) The charging process is simplified to constant power charging. The D_{EV} of private EVs daily trips approximately meets a lognormal distribution (Qian et al., 2011). The starting time of charging t_s not affected by PBDR adopts piecewise

normal distribution (Tian et al., 2010). For EVOs affected by PBDR, the starting time of charging t_s^{PDR} is calculated as follows:

$$t_s^{PDR} = \begin{cases} t_{valley}^{start} + rand \times (t_{valley}^{end} - t_c), & 0 \leq t_c \leq t_{valley}^{end} - t_{valley}^{start} \\ t_{valley}^{start}, & \text{else} \end{cases} \quad (14)$$

where t_{valley}^{start} and t_{valley}^{end} denote the starting and ending time of the valley price. Rand indicates a random number in the range of the interval range in $[0,1]$. The above equations show that when the power consumption time of EVOs is greater than the interval duration of the valley hours, EVOs choose to charge at t_{valley}^{start} , shown in Appendix B.1. When the EVOs charging time is less than the interval duration of the valley hours, the charging time is chosen randomly within the interval. In this way, we can calculate the total charging power per hour. By calculating the charging of each EV through MCS, we can get the total charging power. The EVs load in IES is calculated as $P_k = \sum_{i=1}^N P_i A_{ik}$, where P_i is the charging power of the i -th EV; P_k is the total charging power demand at the k -th hour, A_{ik} is 1 when the i -th EV is charged, otherwise, it is 0, $1 \leq i \leq N$; $1 \leq k \leq 24$. By summing up the charging power of each EV in each time period, we can get the total load of EV charging during the statistical time. Multiple EVs generated by MCS are shown in Figure 22 in Appendix B.3. Stochastic charging demand flow diagram is illustrated in Figure 23 in Appendix B.4. Simulations by these steps, total daily charging load of 1000 EVs including two types of EVOs are shown in Figure 4, and Sensitivity analysis of EV charging load under different electric vehicle group (EVG) and η is seen from Figure 24 in Appendix B.5. .

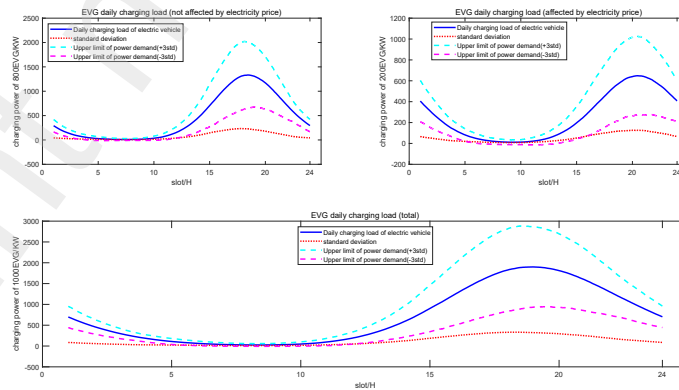


Figure 4 Total daily charging load of EVs including two types of EVOs.

B. Probabilistic photovoltaic model

Solar irradiance r obeys the Beta distribution at a given time period (Y. Li, Han, et al., 2021). Photovoltaic(PV) generation is mainly affected by solar irradiation, ambient temperature and the characteristics of the PV module (Laudani et al., 2014), shown as follows:

$$f_r(r) = \frac{\Gamma(\lambda_1) + \Gamma(\lambda_2)}{\Gamma(\lambda_1)\Gamma(\lambda_2)} \left(\frac{r}{r_{\max}} \right)^{\lambda_1-1} \left(1 - \frac{r}{r_{\max}} \right)^{\lambda_2-1} \quad (15)$$

where the Gamma function $\Gamma(\lambda) = \int_0^{+\infty} p^{\lambda-1} e^{-p} dp$, p is an integer variable, λ_1 and λ_2 are the shape parameters, denotes as $\lambda_1 = u_{pv} \left(\frac{u_{pv}(1-u_{pv})}{\sigma_{pv}^2} - 1 \right)$, $\lambda_2 = (1-u_{pv}) \left(\frac{u_{pv}(1-u_{pv})}{\sigma_{pv}^2} - 1 \right)$. The output PV power P^{PV} is expressed as $P^{PV} = r A^{PV} \eta^{PV}$, where A^{PV} and η^{PV} denotes the radiation area and the conversion efficiency of PV panel. Based on the above relationship, the PDF of PV generation power output denotes below:

$$f_P(P^{PV}) = \frac{\Gamma(\lambda_1) + \Gamma(\lambda_2)}{\Gamma(\lambda_1)\Gamma(\lambda_2)} \left(\frac{P^{PV}}{P_{\max}^{PV}} \right)^{\lambda_1-1} \left(1 - \frac{P^{PV}}{P_{\max}^{PV}} \right)^{\lambda_2-1} \quad (16)$$

where $P_{\max}^{PV}$ represents the maximum PV panel power output.

C. Probabilistic wind turbine model

Previous studies have demonstrated that wind speeds follow the Weibull distribution (Bowden et al., 1983; Wais, 2017), the PDF of the wind speed is $f_w(v) = (k/c)(v/c)^{k-1} \exp[-(v/c)^k]$, where v , k and c are the actual wind speed, the shape factor and the scale factor. The WT output P^{WT} is described using the piecewise function below:

$$P^{WT} = \begin{cases} 0, & v < v_{in}, v \geq v_{out} \\ [(v - v_{in}) / (v_{rated} - v_{in})] P_{rated} & v_{in} \leq v \leq v_{rated} \\ P_{rated} & v_{rated} \leq v \leq v_{out} \end{cases} \quad (17)$$

where v_{in} and v_{out} denote cut-in and cut-out wind speed, v_{rated} and P_{rated} are the rated wind speed and the WT rated power output; $h = (v_{rated} / v_{in}) - 1$. The PDF of the WT generation is denoted below:

$$f_W(P^{WT}) = \begin{cases} 0, & P^{WT} < 0 \\ 1 - \exp\{-(1+hP^{WT}/P_{rated})v_{in}/z\}^w\} & 0 \leq P^{WT} < P_{rated} \\ +\exp[-(v_{out}/z)^w], & \\ 1, & P^{WT} \geq P_{rated} \end{cases} \quad (18)$$

D. Probabilistic load model

Due to the large number of users in IES, according to the Central Limit Theorem, the load fluctuation can be approximated to follow a normal distribution with its PDF given by:

$$f_l(P^L) = \frac{1}{\sqrt{2\pi}\sigma_L} \exp\left(-\frac{(P^L - \mu_L)^2}{2\sigma_L^2}\right) \quad (19)$$

where P^L is the fluctuating load (Y. Li et al., 2022).

2.7. Subsystem model

A. CCHP model

Combined cooling heating and power (CCHP) is a subsystem that generates electricity, heat and cold energy simultaneously through the combustion of fuels such as natural gas, which consists of three components: conventional gas turbine (CGT), waste heat boiler (WHB), and absorption chiller (AC). The overall efficiency for CCHP subsystem is 70% to 85% for district heating and cooling. The mathematical expressed below:

$$\begin{cases} Q_{wh} = H_{whb} + H_{ac} \\ Q_{ac} = \gamma_{ac} H_{ac} \\ Q_{whb} = \gamma_{whb} H_{whb} \end{cases} \quad (20)$$

The balance of the cold and thermal loads are shown as follows:

$$L_c + C_c^{st} = P_c^{net} + Q_{ac} + Q_{ec} \quad (21)$$

$$L_h + C_h^{st} = P_h^{net} + Q_{whb} + Q_{gb} + Q_{eb} \quad (22)$$

B. HESS model

The hybrid energy storage system (HESS) contains electrical storage device, thermal storage device, cooling storage device and gas storage tank. All conform to the following mathematical model:

$$C_{t+1} = (1 - k_{loss})C_t + (\eta_{ch}P_{ch}^t - P_{dc}^t / \eta_{dc})\Delta t \quad (23)$$

C. P2G model

The power to gas (P2G) technology involves two reaction processes. H_2O is electrolyzed into H_2 and O_2 using an EL, and then CH_4 is produced from H_2 and CO_2 using methane reactor. In the second process, CO_2 can be captured from waste gas and biogas by carbon capture systems, or directly from the air using direct air capture technology. The model is denoted below:

$$Q_i^{P2G} = \frac{\eta^{P2G} P_i^{P2G}}{HHV} \quad (24)$$

D. CGT model

The input gas to the Conventional gas turbine (CGT) of CCHP is formulated as follows:

$$Q_{gas} = \frac{P_{gt}(t)T(t)}{\beta_1 * HHV} + \frac{Q_{gb}}{\beta_2 * HHV} \quad (25)$$

where Q_{gb} is the thermal load provided by GB; P_{gt} is the output power of CGT; β_1 and β_2 are the power generation efficiency of CGT and the heating efficiency of GB.

The waste heat is generated by the CGT with the following expression:

$$Q_{wh} = P_{GT}(t) \frac{1 - \beta_1 - \varepsilon}{\beta_1} \quad (26)$$

2.8. The optimization operation model of IES

2.8.1. Objective function

The objective function constructed in IES includes both economic and environmental aspects, and the objective of this IES is to minimize daily comprehensive cost (DCC) and daily carbon emissions (DCE). Since economic benefits are often achieved at the cost of more carbon emissions, these two objective functions are contradictory. The first objective expresses the economic performance of IES through the indicator of DCC; the second objective reflects the environmental performance of IES through the indicator of DCE. We first perform the optimization solution for each objective separately, and the optimization results are labeled as DCC_1 and DCE_1 . Then, a multi-objective function is transformed into a single objective optimization problem, which is constructed to optimize DCC and DCE based on the decision maker's preference, thus, the objective function of the above model is written as below:

$$\min F = W_1 \frac{DCC_2 - DCC_1}{DCC_1} + W_2 \frac{DCE_2 - DCE_1}{DCE_1} \quad (27)$$

where F denotes the deviation preference of decision-makers, W_1 and W_2 indicate the decision maker's preference of cost and environment, $W_1 + W_2 = 1$.

A. Daily comprehensive cost

Considering the energy purchase cost of IES, the LTCT cost and the wind abandonment cost, the low-carbon economic scheduling objective is constructed to minimize DCC, described as follows:

$$DCC_1 = \min(f_{om} + f_{trans} + f_{DG}^{cut} + f_{CO_2} + f_{envir} + f_{reserve}) \quad (28)$$

$$f_{om} = \sum_{t=1}^T \sum_{l=1}^L P_{l,t} \omega_{l,om} \quad (29)$$

$$f_{trans} = \sum_{t=1}^T \left(\omega_{e,t}^{P,buy} P_{e,t}^{P,buy} - \omega_{e,t}^{P,sell} P_{e,t}^{P,sell} + \omega_{g,t}^{G,buy} P_{g,t}^{G,buy} - \omega_{g,t}^{G,sell} P_{g,t}^{G,sell} + \right. \\ \left. \omega_{h,t}^{H,buy} P_{h,t}^{H,buy} - \omega_{h,t}^{H,sell} P_{h,t}^{H,sell} + \omega_{c,t}^{C,buy} P_{c,t}^{C,buy} - \omega_{c,t}^{C,sell} P_{c,t}^{C,sell} \right) \quad (30)$$

$$f_{DG}^{cut} = \delta_{DG} \sum_{t=1}^T P_{DG}^{cut}(t) \quad (31)$$

$$f_{envir} = \sum_{t=1}^T \sum_{j=1}^J \phi_j \left[\mu_{p,j}^P (P_{e,t}^{P,buy} - P_{e,t}^{P,sell}) + \mu_{g,j}^G (G_{g,t}^{G,buy} - G_{g,t}^{G,sell}) HHV \right. \\ \left. + \mu_{h,j}^H (P_{h,t}^{H,buy} - P_{h,t}^{H,sell}) + \mu_{c,j}^C (P_{c,t}^{C,buy} - P_{c,t}^{C,sell}) \right] \quad (32)$$

$$f_{reserve} = \sum_{t=1}^T (\delta_t^{grid} R_t^{grid} + \delta_t^{ESD} R_t^{ESD}) \quad (33)$$

where the LTCT cost f_{CO_2} is seen in Eq.(8), P_t and ω_t are the output power and energy price at time slot t . δ_{DG} is the unit penalty cost of wind abandonment; $f_{DG}^{cut}(t)$, f_{envir} and $f_{reserve}$ is the abandoned wind power, environmental cost and reserve spinning cost at time slot t , HHV is the Calorific value of natural gas.

B. Daily carbon emission

The main carbon emissions of IES come from natural gas. The sources of CO₂ emissions from the IES are GTs and GBs.

$$DCE = Q_{gas} \phi_1 \quad (34)$$

$$Q_{gas} = \frac{P_{GT}(t)T(t)}{\beta_1 \delta} + \frac{Q_{nb}}{\beta_2 \delta} \quad (35)$$

2.8.2. Constraints

A. Energy balance constraints

To avoid supply and demand imbalance by excessive output of renewable energy power generation (REPG), IES is equipped with a load adjustment load-shedding P^s with the following power balance constraints for electricity, heat, cold, gas and hydrogen:

$$P_{e,t}^{net} + P_{e,t}^{pv} + P_{e,t}^{wt} + P_{e,t}^{CCHP} + P_{e,t}^{HFC} = P_{e,t}^E + P_{e,t}^{EL} + \sum_{n=1}^N P_{n,t}^{EV} + P_t^{ESD} + P_{e,t}^{P2G} + P_{e,t}^s \quad (36)$$

$$P_{h,t}^{net} + P_{h,t}^{GB} + P_{h,t}^{CCHP} + P_{h,t}^{HFC} = P_{h,t}^{HL} + P_t^{HST} + P_{h,t}^s \quad (37)$$

$$P_{c,t}^{net} + P_{c,t}^{CCHP} = P_{c,t}^{CL} + P_t^{CST} + P_{c,t}^s \quad (38)$$

$$P_{g,t}^{net} + P_{g,t}^{MR} = P_{g,t}^{GL} + P_{g,t}^{GB} + P_t^{GST} + P_{g,t}^{CCHP} + P_{g,t}^s \quad (39)$$

$$P_{h2,t}^{net} + P_{h2,t}^{EL} = P_{h2,t}^{MR} + P_{h2,t}^{HFC} + P_t^{GST} + P_{h2,t}^s \quad (40)$$

B. Storage device constraints

The modeling of electric, thermal, gas, and hydrogen storage devices is denoted below:

$$\begin{cases} 0 \leq P_{ES,n}^{cha}(t) \leq B_{ES,n}^{cha}(t) P_{ES,n}^{max} \\ 0 \leq P_{ES,n}^{dis}(t) \leq B_{ES,n}^{dis}(t) P_{ES,n}^{max} \\ B_{ES,n}^{cha}(t) + B_{ES,n}^{dis}(t) = 1 \\ P_{ES,n}(t) = P_{ES,n}^{cha}(t) \eta_{ES,n}^{cha} - P_{ES,n}^{dis}(t) / \eta_{ES,n}^{dis} \\ S_n(t) = S_n(t-1) + P_{ES,n}(t) / P_{ES,n}^{cap} \\ S_n(1) = S_n(T) \\ S_n^{min} \leq S_n(t) \leq S_n^{max} \end{cases} \quad (41)$$

C. External grid constraints

The main grid constructed by IES is a grid-connected and interactive system that not only buys energy from the main grid as a demand side, but also sells energy to it as a supply side. The energy interaction process must satisfy the following conditions:

$$P_t^{buy} + P_t^{sell} = P_t^{net} \quad (42)$$

$$\begin{cases} P_t^{net} \geq 0, P_t^{buy} = P_t^{net}, P_t^{sell} = 0, & \text{if } \zeta_{emp}^{net} = 1 \\ P_t^{net} \leq 0, P_t^{sell} = P_t^{net}, P_t^{buy} = 0, & \text{if } \zeta_{emp}^{net} = 0 \end{cases} \quad (43)$$

where ς_{emp}^{net} denotes the interactive power sign.

D. Power operating constraint

(1) Capacity restriction of all devices:

$$\begin{aligned} \mu_t P_t^{k,\min} &\leq P_{t,t}^k \leq \mu_t P_t^{k,\max}, \forall k \in \text{all devices} \\ \mu_t &= \begin{cases} 1, & \text{if the unit is on at time slot } t \\ 0, & \text{if when the unit is off at time slot } t \end{cases} \end{aligned} \quad (44)$$

where $P_t^{\min}$ and $P_t^{\max}$ is the minimum and maximum power output, μ_t is the start-up or shut-down of all devices.

(2) The start-up and shut-down constraints:

$$\begin{cases} (\mu_{t-1} - \mu_t)(T_{t-1}^{on} - T^{on}) \geq 0 \\ (\mu_t - \mu_{t-1})(T_{t-1}^{off} - T^{off}) \geq 0 \end{cases} \quad (45)$$

where T_{t-1}^{on} and T_{t-1}^{off} denote the continuous on/off time at the last time slot t . The devices in operation cannot remain in a continuous period of on and off states.

(3) Climbing power constraint:

$$\begin{cases} \Delta P_{down} \leq P_t - P_{t-1} \leq \Delta P_{up} \\ 0 \leq P_t \leq P_{t,rated} \end{cases} \quad (46)$$

where ΔP_{down} and ΔP_{up} are the falling power and climbing power; $P_{t,rated}$ is the rated power.

E. Electric vehicles constraints

The total EVs charging power cannot exceed the maximum charging capacity of the EV charging station, at the same time, the initial EV capacity of each cycle is equal to the EV capacity of the end time period within this cycle, and the individual EV capacity is affected by the battery capacity of the last time slot, the charging power and charging efficiency, which must meet the following conditions:

$$\begin{cases} P_{ch,\min}^{EV} \leq \sum_{n=1}^N P_{ch,n,t}^{EV} \leq P_{ch,\max}^{EV} \\ C_n^{EV}(0) = C_n^{EV}(24) \\ C_{n,t}^{EV} = C_{n,t-1}^{EV} + P_{ch,n}^{EV} \eta_{ch}^{EV} \forall t \\ C_{\min,n}^{EV} \leq C_{n,t}^{EV} \leq C_{\max,n}^{EV} \end{cases} \quad (47)$$

F. Spinning reserve constraints

ESD and the grid participate in the function of providing reserve services through spinning reserve units. In extreme cases where sufficient renewable energy output is not enough, the spinning reserve will perform the function of providing energy. The constraints of spinning reserve are expressed as a chance constraint form (Y. Li et al., 2019):

$$R_t^{grid} \leq P_{\max}^{e,grid} - (P_{e,t}^{EL} + P_{e,t}^{H2L} + P_{e,t}^{CL} + P_{e,t}^{GL} + P_{e,t}^{HL}) \quad (48)$$

$$R_t^{ESD} \leq \min \left\{ \eta_{dc} (C_t^{ESD} - C_{\min}^{ESD}) / \Delta t, P_{dc,\max}^{ESD} - P_{dc,t}^{ESD} \right\} \quad (49)$$

$$P_{rob} \left\{ R_t^{grid} + R_t^{ESD} \geq E(P_t^{RG}) - P_t^{WT} - P_t^{PV} \right\} \geq \alpha \quad (50)$$

where α is confidence level participate in the process of spinning reserve (Y. Li, Wang, et al., 2021).

2.9. Analytical framework and solution approach

In this section, our main purpose is to transform the original probabilistic model into a deterministic solvable MILP.

A. The probabilistic sequence

Sequence operation theory (SOT) is a new general mathematical theory and widely used in the reliability of power system (Kang et al., 2008), which was proved to be a good tool to transform the chance-constrained programming (CCP) of spinning reserve into its deterministic equivalence form (Kang et al., 2008). Meanwhile, to the best of our knowledge, discretized step transformation (DST) is a good mathematical tool in dealing with multiple uncertainties based on SOT (Y. Li, Han, et al., 2021). As the natural properties of sunlight and wind, PV and WT generation are random and complementary, based on the above theories, we can theoretically handle the uncertainties of REPG by discretizing the PDF of PV and WT into many pieces to get the probability sequences $a(i_{at})$ and $b(i_{bt})$ by the concept derived from calculus. The probabilistic sequence length N_{at} and N_{bt} denote below:

$$\begin{cases} N_{at} = \lceil P_{\max,t}^{PV} / q \rceil \\ N_{bt} = \lceil P_{\max,t}^{WT} / q \rceil \end{cases} \quad (51)$$

where $\lceil g \rceil$ denotes the ceiling function; $P_{max,t}^{PV}$ and $P_{max,t}^{WT}$ is the maximum output power of the PV and WT unit at time slot t ; q is the discrete step size.

According to SOT, the probabilistic sequence $a(i_{at})$ of the PV output (Y. Li et al., 2020) is given below:

$$a(i_{at}) = \begin{cases} \int_0^{q/2} f_p(P^{PV}) dP^{PV}, & i_{at} = 0 \\ \int_{i_{at}q - q/2}^{i_{at}q + q/2} f_p(P^{PV}) dP^{PV}, & 0 < i_{at} < N_{at} \\ \int_{i_{at}q - q/2}^{i_{at}q} f_p(P^{PV}) dP^{PV}, & i_{at} = N_{at} \end{cases} \quad (52)$$

Similarly, the probabilistic sequence $b(i_{bt})$ of WT outputs is formulated below:

$$b(i_{bt}) = \begin{cases} \int_0^{q/2} f_p(P^{WT}) dP^{WT}, & i_{bt} = 0 \\ \int_{i_{bt}q - q/2}^{i_{bt}q + q/2} f_p(P^{WT}) dP^{WT}, & 0 < i_{bt} < N_{bt} \\ \int_{i_{bt}q - q/2}^{i_{bt}q} f_p(P^{WT}) dP^{WT}, & i_{bt} = N_{bt} \end{cases} \quad (53)$$

Therefore, the probabilistic sequence of the joint generation outputs $c(i_{ct})$ is derived:

$$c(i_{ct}) = \sum_{i_{at} + i_{bt} = i_{ct}} a(i_{at}) b(i_{bt}), \quad i_{ct} = 0, 1, \dots, N_{at} + N_{bt} \quad (54)$$

The expectation of REPG $E(P_t^{REPG})$ is gained as follows:

$$\begin{aligned} E(P_t^{REPG}) &= E(P_t^{PV}) + E(P_t^{WT}) \\ &= \sum_{m_{at}=0}^{N_{at}} m_{at} q a(m_{at}) + \sum_{m_{bt}=0}^{N_{bt}} m_{bt} q b(m_{bt}) \end{aligned} \quad (55)$$

B. Reformulation of CCP

To convert the CCP (50) to the deterministic equivalent form, hereby we denote $Z = P_t^{WT} + P_t^{PV}$, the cumulative distribution function of the variable Z is below:

$$F_Z(z) = \int_{-\infty}^z \left[\int_{-\infty}^{\infty} f_w(z-y) f_p(y) dy \right] dz \quad (56)$$

As the above PDF_s is difficult to solve, we adopt SOT to solve the probability distribution of the variable Z by converting the CCP to its deterministic form. The probability sequence of the joint output $c(m_{ct}) = \sum_{m_{at}+m_{bt}=m_{ct}} a(i_{at})b(m_{bt})$, $m_{ct} = 0, 1, 2, \dots, N_{at} + N_{bt}$, and the joint REPG probability sequence is listed in Table 2.

Table 2 The joint REPG probabilistic sequence.

Power (MW)	0	q	...	$m_c q$	...	$N_{ct} q$
Probability	$c(0)$	$c(1)$	...	$c(m_c)$	...	$c(N_{ct})$

ω_m is hereby introduced to solve the Eq.(50), expressed below:

$$\omega_m = \begin{cases} 1, R_t^{grid} + R_t^{ESD} \geq E(P_t^{RG}) - m_{ct}q, m_{ct} = 0, 1, \dots, N_{ct} \\ 0, \text{else} \end{cases} \quad (57)$$

To convert the eq.(50) into a deterministic equivalent form, CCP is simply converted into the following form:

$$\sum_{m_{ct}=0}^{N_{at}+N_{bt}} \omega_m c(m_{ct}) \geq \alpha \quad (58)$$

C. Linearization process

A binary variable $Z_{m_{ct}}$ is introduced to solve this nonlinear programming problem, denotes as follows:

$$\frac{\left[R_t^{grid} + R_t^{ESD} + m_{ct}q - E(P_t^{RG}) \right]}{M} \leq Z_{m_{ct}} \leq \frac{\left[R_t^{grid} + R_t^{ESD} + m_{ct}q - E(P_t^{RG}) \right]}{1 + \frac{M}{M}} \quad (59)$$

$\forall t, m_{ct} = 0, 1, \dots, N_{at} + N_{bt}$

where M is large enough to be a positive number, $Z_{m_{ct}}$ equal to 1 or 0, $R_t^{grid} + R_t^{ESD} \geq E(P_t^{RG}) - m_{ct}q$.

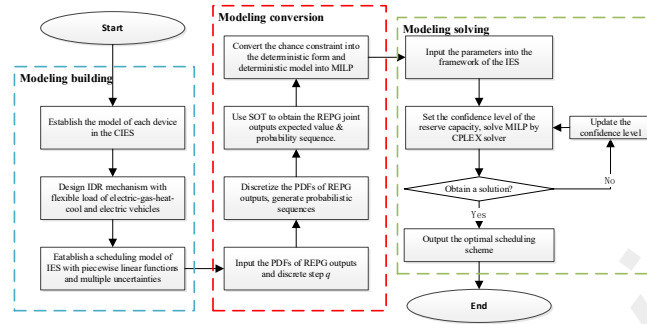


Figure 5 Flow chart of the proposed solution.

By converting the chance-constrained programming (CCP) into a deterministic equivalent form, and then this equivalent form is a solvable mixed integer linear programming (MILP). The solution procedure of the proposed DST method is shown in Figure 5.

3. Simulation

3.1. Description of the proposed IES

The main parts of this proposed IES consists of REPG, CCHP, HESS, AUES subsystem and a carbon trading market, where PV and WT comprise REPG subsystem, CGT, WHB, AC make up CCHP subsystem, BAT, GST, HST, H₂ST, CST form the HESS system, and GB, HFC, MR, EL constitutes AUES subsystem. In particular, the carbon trading market in IES is designed by carbon quota and LTCTM, users can freely trade carbon tax like trading commodities according to their actual needs. AUES plays an auxiliary role for the safe and stable operation, which can provide energy conversion during special emergency periods to ensure the smooth operation of IES. All the data and parameters in this simulation can be found in (Clegg & Mancarella, 2015; Y. Li, Han, et al., 2021; Zhou et al., 2021; Zhu et al., 2023) , and the environmental parameters in (32) are from (Clegg & Mancarella, 2015).

3.2. Results of IES optimal scheduling

From Figure 6, we can see that the optimization of IES operation achieves a load balance of energy input and output in a typical day for different types of energy, the electric heat ratio and the electric cooling

ratio of the adjustable CCHP are illustrated in Figure 7 with the electric heat ratio of the HFC remaining constant. SOC of various types of energy storage devices is shown in Figure 8.

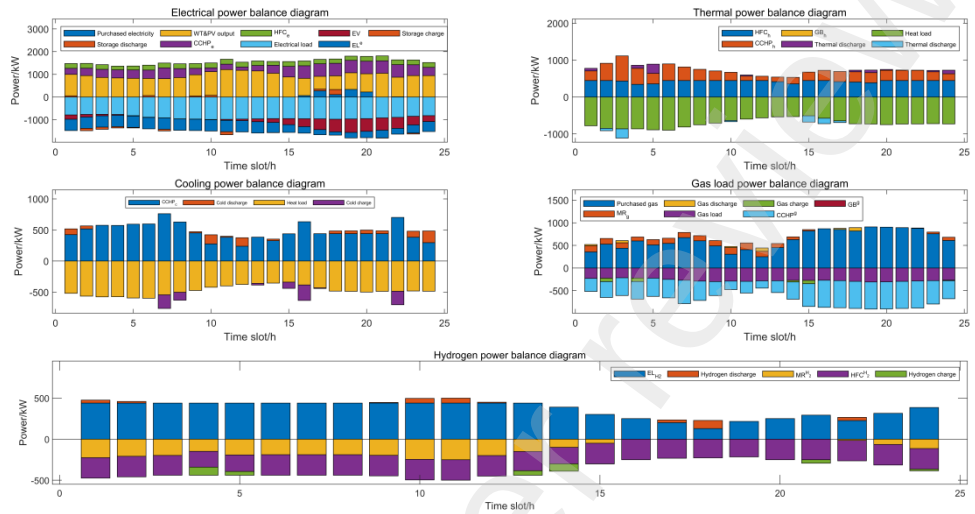


Figure 6 Typical daily load supply for different devices.

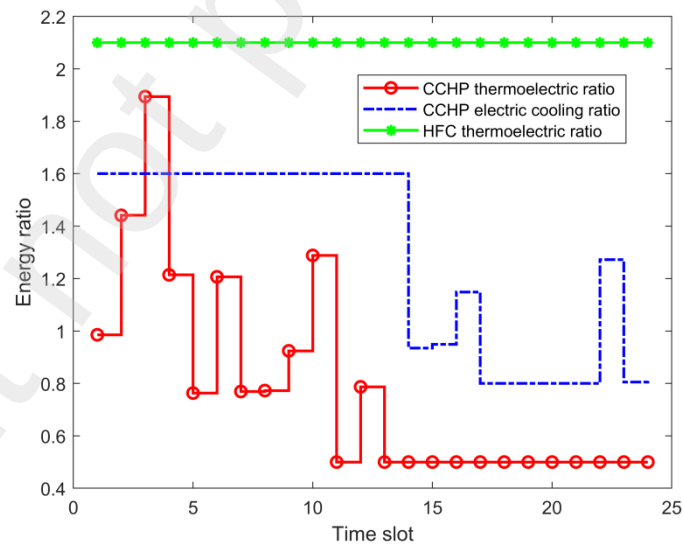


Figure 7 The energy ratio.

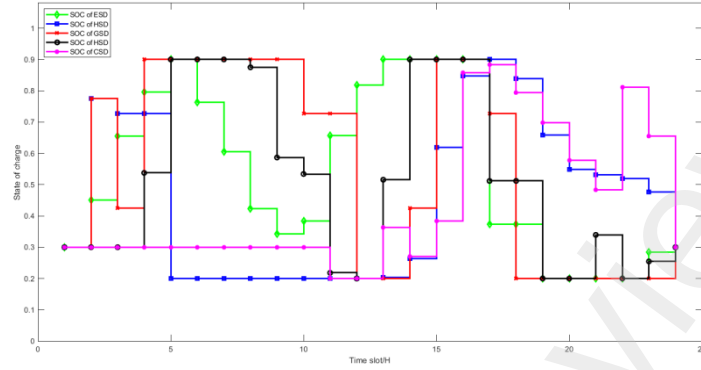


Figure 8 SOC of various types of energy storage devices.

3.3. Discussion of the RTP model

The benchmark parameters of this IES on the welfare model are as follows: the running cycle one day is $K = 24$ hours, the number of users in this IES is $N = 12$, the piece number of piecewise function $P = 16$, the elastic coefficient is $\omega = rand(3,4)$, the smooth coefficient $\varepsilon = 0.01$, the utility function coefficient is $\alpha = 0.25$, the boundary of the electricity price $\lambda \in [\lambda_{\min}, \lambda_{\max}] = [0.2, 0.9]$, the Interval of the per user's electricity consumption $\gamma \in [\gamma_{\min}, \gamma_{\max}] = [1, 8]$, the coefficient of the cost function is $a = 0.01, b = 0, c = 0$. Accordingly, the RTP, total utility and total electricity consumption can be obtained.

3.4. RTP model under different parameters

Based on the theory of Section 2.1 and Section 2.2, we characterize the utility and cost functions by PWLF, and then using the smoothing method to make the social welfare model into a solvable convex optimization problem. In addition, we can obtain the RTP by the Lagrange multiplier using the dual method.

From Figure 9, we find that the number of users has a large impact on the price, and the more the number of pieces, the more stable the electricity price. From Figure 10, the overall electricity consumption of users in each time slot is less affected by parameter changes, and the electricity consumption is very close to the generation, with high electric utilization. From Figure 11, the number of users has little effect on the total utility, and when the number of pieces exceeds 8, the greater the number of pieces, the greater the effect of the number of pieces

on the total utility. We have performed numerical simulations of a social welfare model considering PWLFs, with sensitivity analysis for different parameters in terms of the number of different pieces users, respectively.

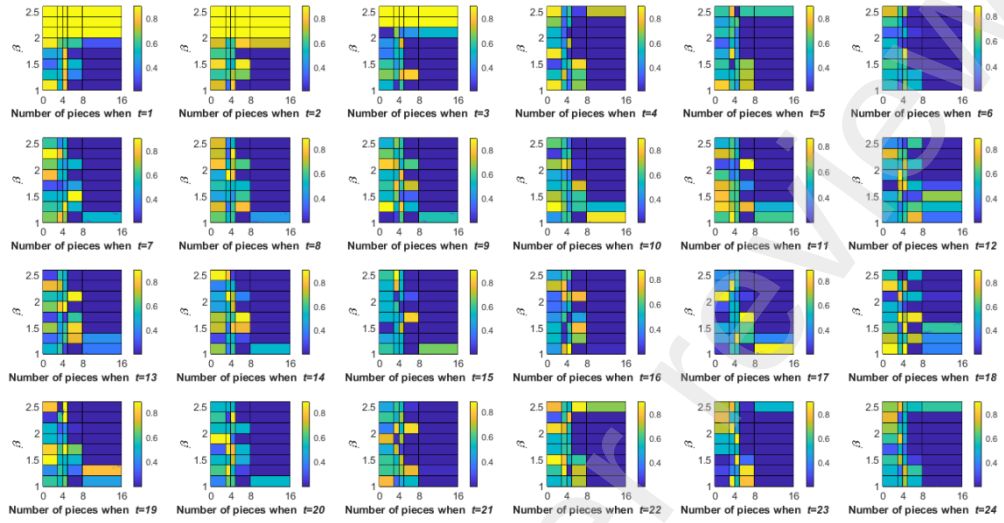


Figure 9 RTP under different number of users and pieces.

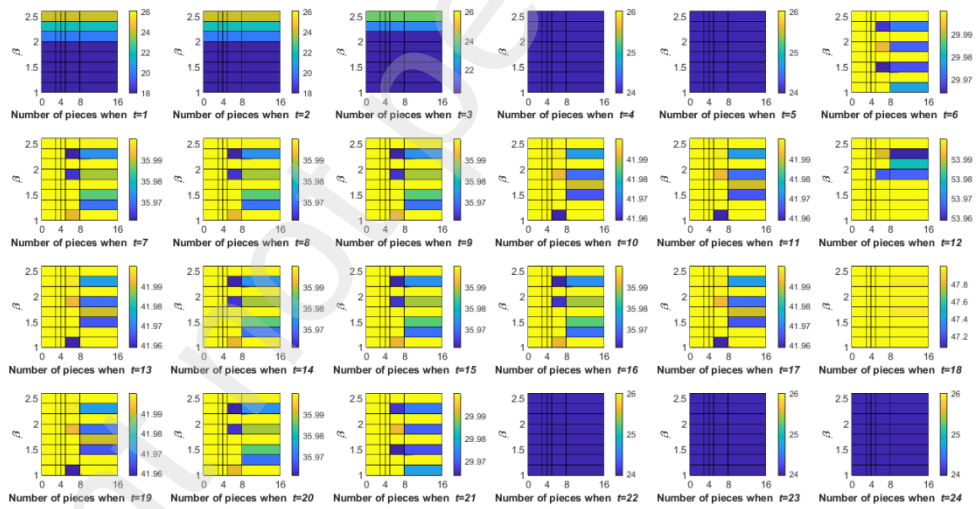


Figure 10 Loads under different number of users and pieces.

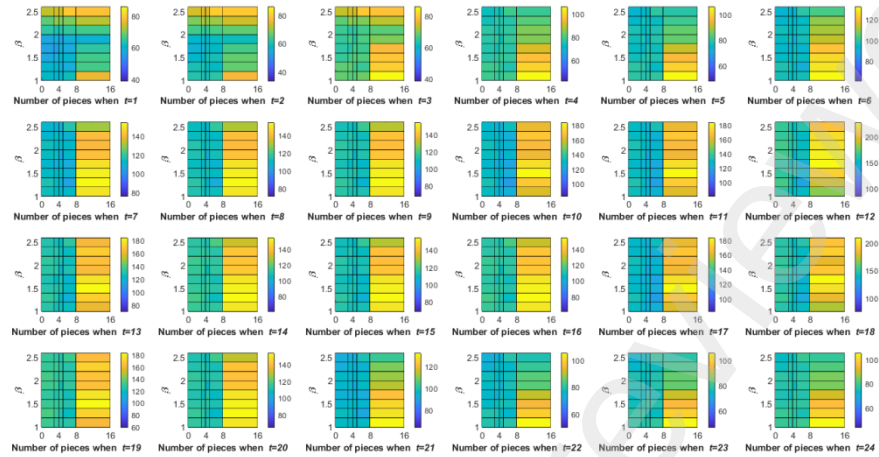


Figure 11 Welfare under different number of users and pieces.

A. Different piece function model

Figure 12 shows the simulation results of the electricity price for each piece function model, the electricity price under the quadratic, three-piece, four-piece and five-piece function models fluctuates relatively larger, while the electricity price under the eight-piece and sixteen-piece function models fluctuates slightly and tends to be stable. Moreover, with the increase of the number of pieces, the electricity price of the piecewise function model tends to be flat.

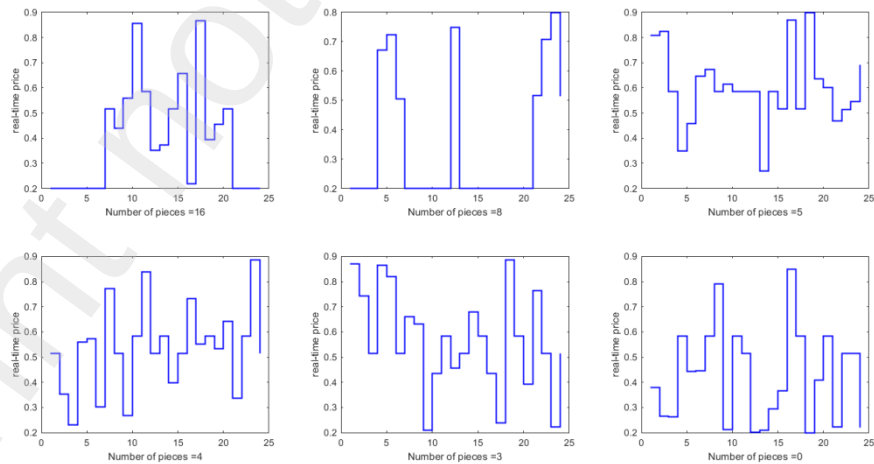


Figure 12 RTP under different number of pieces.

Figure 13 shows the simulation results of the welfare for each piece function model, the total utility under the quadratic function model fluctuates between 30 to 90, and three to eight piece function models fluctuates between 60 to 180, while the sixteen-piece function models fluctuate between 80 to 200, which indicates that the higher the number of pieces under the function model, the higher the satisfaction of users and providers.

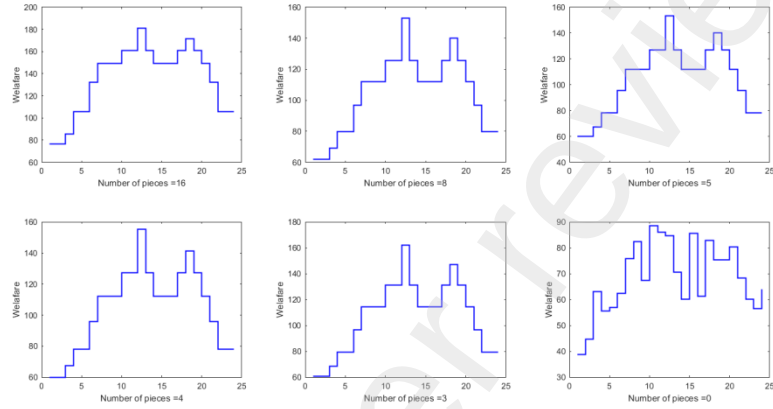


Figure 13 Welfare under different number of pieces.

Figure 14 shows the simulation results of the total power for each piece function model, there is no difference in the fluctuations of the total electricity consumption under the different piece function model, and the electricity consumption all fluctuates between 15 and 55, reaching the first peak at around 12:00 a.m. and the second peak at around 18:00 p.m., which is consistent with the actual electricity consumption of users.

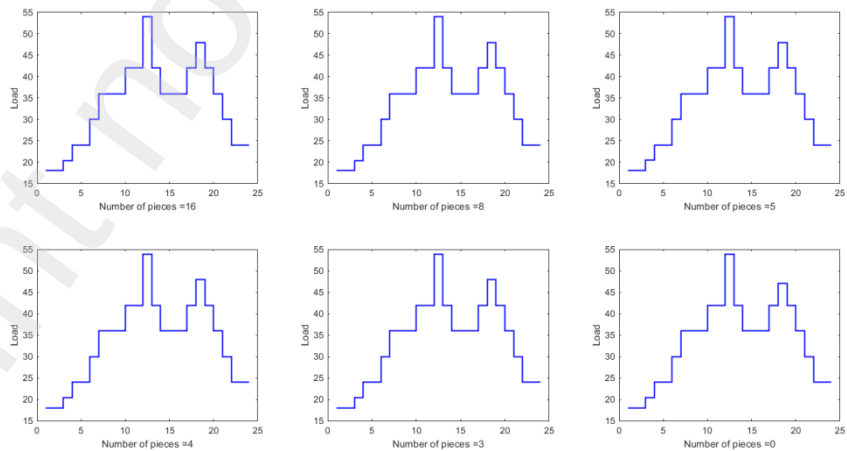


Figure 14 Loads under different number of pieces.

In general, as the number of pieces increases, the electricity price tends to be stable, and the total utility of both users and power providers increases simultaneously. The welfare model is strongly affected by these parameters.

B. Different number of users function model

Figure 15 shows the price simulation results of the sixteen-piece function model for different users. The price fluctuates between 0.2 and 0.9 under 10 users, and as the number of users increases, the price tends to stabilize the upper or lower price boundry at 0.2 or 0.9. The electricity price tends to stabilize when the number of users exceeds 20 ($\beta = 2$), this is because when the number of users reaches a certain level, according to the law of large numbers, the overall electricity consumption of users tends to be stable and the price tends to be flat. Figure 16 shows the welfare simulation results of the sixteen-piece function model for different users. The total utility fluctuates within 250, with two peak periods, the first one at around 12:00 a.m. and the second one at about 18:00 p.m. Figure 17 shows the loads simulation results of the sixteen-piece function model for different users. The total electricity consumption fluctuates between 10 and 60, and as the number of users increases, the total electricity consumption also increases, which coincides with the trend of the total utility.

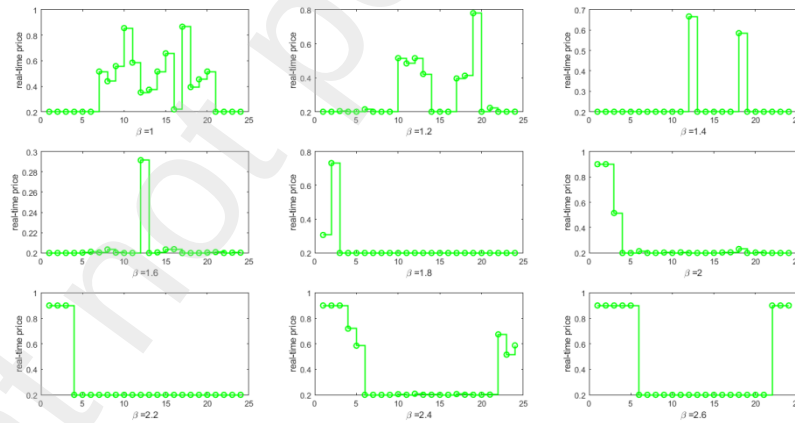


Figure 15 RTP under different number of users

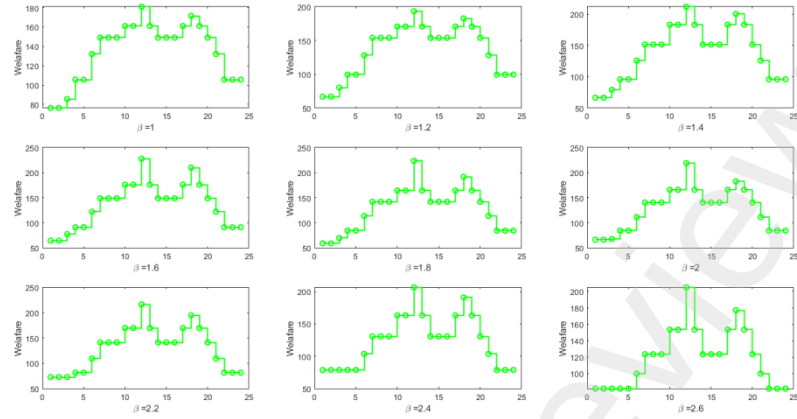


Figure 16 Welfare under different number of users.

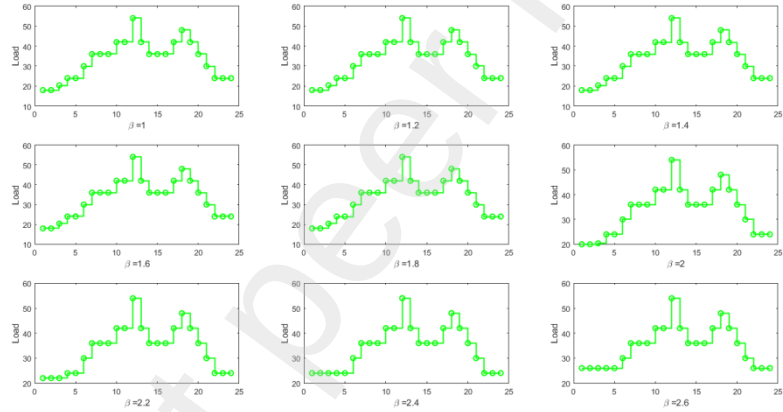


Figure 17 Loads under different number of users.

In general, as the number of pieces and users increases, the corresponding RTP, welfare, and electricity loads become increasingly smooth, while the welfare level becomes higher. The PWLF model is somewhat roughly portrayed, while the total utility and electricity price of the function model are smoother as the number of pieces increases.

4. Further discussion and analysis

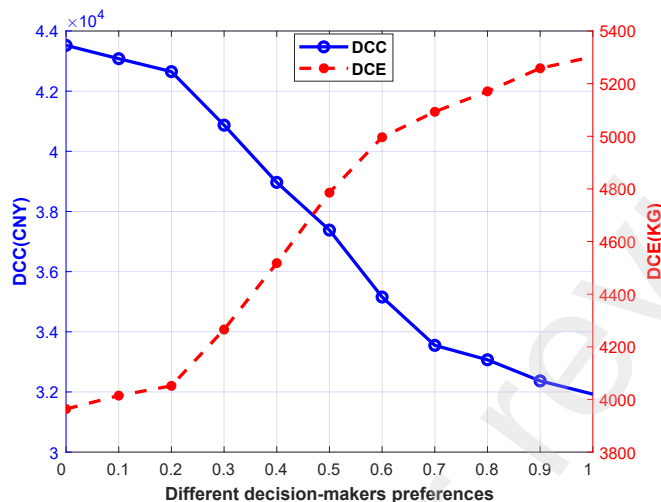


Figure 18 DCC and DCE under different optimization preferences.

4.1. Sensitivity analysis

As for the objectives of DCC and DCE, we can see the different optimization preferences results are shown in Figure 18. In this IES, there is a negative correlation in DCE and DCC, as DCC increases, DCE decreases. When decision preference is shifted from $(W_1 = 0.1, W_2 = 0.9)$ to $(W_1 = 0.2, W_2 = 0.8)$, the change in decision makers' preferences has little effect on DCE and cost fluctuations, while when preferences are shifted from $(W_1 = 1, W_2 = 0)$ to $(W_1 = 0.7, W_2 = 0.3)$, the lower cost increases yield relatively limited carbon dioxide emissions reduction (CDER), and it indicates that the optimal shift is not cost-effective enough. Therefore, from Figure 18, the result shows that it is unsuitable to pursue environmental goals and cost objectives in a separate way, and the optimal range of optimization preferences for decision-makers focuses on the interval from $(W_1 = 0.2, W_2 = 0.8)$ to $(W_1 = 0.7, W_2 = 0.3)$.

4.2. Carbon trading mechanisms analysis

Most of the existing literature only studies the impact of carbon trading base prices on the system, and little literature analyzes the impact of interval step size, carbon tax price growth rate, and the number of

pieces on the system operation. To further study the impact of different parameters of the carbon trading mechanism on the operation of IES, we discuss the impact of parameters on the total cost of system operation and carbon emissions, shown in Figure 19.

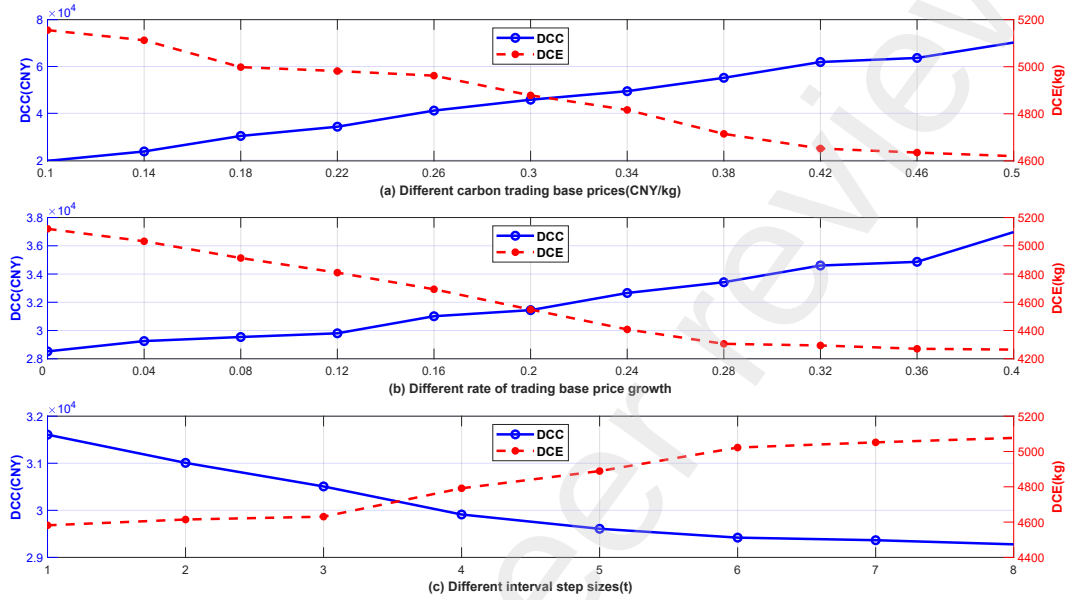


Figure 19 DCC and DCE under different carbon trading mechanism parameters.

As shown in Figure 19 (a), when the carbon trading base price is less than 420 CNY/kg, with the increase of carbon trading base price, the carbon trading cost becomes higher, and IES has to cut carbon emissions to reduce the carbon trading cost, so DCE gradually decreases; while when the carbon trading base price is more than 420 CNY/kg, with the increase of carbon trading base price, the output distribution of devices in IES tends to be limited and DCE also tends to be stable, so the carbon emission is less affected by the change of carbon trading base price. As the carbon trading cost increases, DCC also increases.

As seen in Figure 19 (b), when the growth rate of carbon trading base price varies from 0 to 28%, the carbon trading cost increases as the price growth rate increases, and the IES adjusts the output distribution of devices to reduce DCE for reducing carbon trading cost. Meanwhile, due to the inherent load demand within IES, when the price growth rate exceeds 28%, the devices output distribution tends to be limited and the carbon emission changes also tend to be stable. As the price growth rate increases, the carbon trading cost increases, and DCC in this IES becomes larger.

As shown in Figure 19 (c), when the interval step length is in (1,3], the IES will purchase carbon quotas at a ladder-type price due to the small step length, the carbon transaction cost is higher and therefore DCE is smaller. When the interval length varies in (3,6], as the step length increases due to the inherent load demand within the IES, the smaller the amount of carbon emission quotas that need to be purchased at the higher ladder-type price, and therefore the smaller the carbon transaction cost. As the carbon emission cost gradually decreases, DCE gradually increases. When the interval length varies in (6,8], DCE is traded by the base price and the lower ladder-type price, and the interval length has fewer effects on DCE, so DCE reaches a stable level, as the interval length gradually increases, the carbon transaction cost of the IES gradually decreases, and DCC also gradually decreases.

4.3. Comparative analysis

To analyze the operating levels of different EVGs in IES and non-IES regions, we construct a comparative calculation for both cases using DCE and DCC. Figure 20(a) shows the DCC and DCE for different EVG sizes, and overall, they show an approximately linear increase as the number of EVs increases, and Figure 20(b) shows the comparison between IES and non-IES regions for different sizes of EVG under the associated incremental charging costs and incremental carbon emissions are compared.

Assumptions are given below: (i) DCE in IES area is directly generated by gas from the external grid. (ii) The total EVG charging cost includes charging cost and charging service fee, both of which are 0.8 ¥/kWh, generated by charging pile investment, operation and maintenance; (iii) The charging cost of non-IES consists of time-of-use price and charging service fee, in which the charging electricity is generated by thermal power generation, and the EVG charging cost of IES is based on the case of no EV charging demand, and after considering a certain scale of EV, the incremental DCC of the IES is used as the charging cost, plus the same charging service charge in non-IES area, all of which are used as EVG charging cost.

In general, the charging cost of IES is higher than that of non-IES area, and the charging in this IES is not cost-effective enough till now, but in terms of DCE, they are comparable, and the carbon emission of charging in IES is slightly higher when the number of EVs is less than 200, but with the increase of EV charging scale, the carbon emission of charging in IES is lower and has more environmental advantages. In addition, with the development of

REPG, the unit generation cost of PV and WT will become lower and lower, so charging EVs in IES area will be economical in the near future.

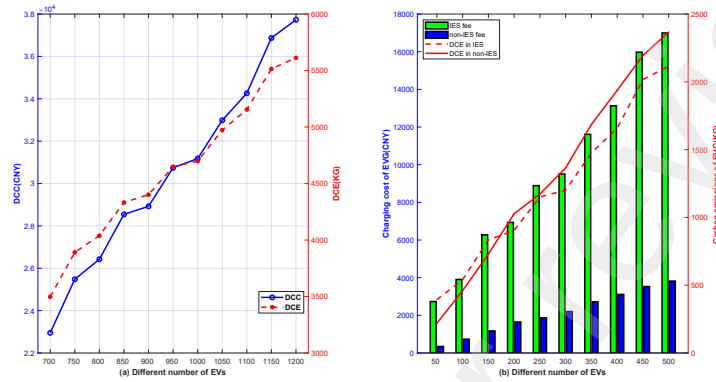


Figure 20 Charging comparison of different EVG between IES and non-IES.

5. Conclusions

In this paper, a novel IES was proposed that takes into account multiple uncertainties as well as a LTCTM to balance the economy and low carbon in a RTP environment. Compared with previous studies on IES scheduling optimization mainly on the supply side, our model considers the impact of various energy sources in real-time price in smart grids and performs systematic optimization and analysis based on the LTCTM to integrate multiple energy forms and load demands. The main conclusions are summarized as follows:

(1) Considering the multiple uncertainties of IES for REPG in the form of joint PDF distributions, the spinning reserve provided by the ESD and the external grid is built in the form of CCP, and by setting the proper confidence level, the IES operation can achieve a trade-off between economy and reliability.

(2) Sensitivity analysis is used to derive the scheduling optimization results under different decision preferences, and the result shows that the optimal optimization ranges between ($W_1 = 0.2, W_2 = 0.8$) and ($W_1 = 0.7, W_2 = 0.3$).

(3) When comparing EVs in IES and non-IES, although EV charging in IES is not currently economically attractive, in terms of CDER, the environmental advantages of IES start to emerge when the number of EVs exceeds 200.

In future studies, the impact of PBDR mechanism on IES scheduling model through RTP and vehicle to grid (V2G) will be further considered, as well as the interaction between the external grid and IES, the impact of electricity consumption and the interaction with the external grid to adapt to the energy demand in the new smart grid environment.

A.1. The mathematical derivation of the error $\left| \hat{f}_{\varepsilon_i}(x) - f_i(x) \right|$ is listed below:

1) When $0 < x \leq a_1$,

$$\begin{aligned} \left| \hat{f}_{\varepsilon_1}(x) - f_1(x) \right| &= \left| \frac{(\omega_1 + \omega_2)x + (\omega_1 - \omega_2)a_1}{2} + \frac{(x - a_1)^2(\omega_2 - \omega_1)}{2((x - a_1)^2 + \varepsilon)^{\frac{1}{2}}} - \omega_1 x \right| \\ &= \frac{1}{2} \left| \frac{(x - a_1)(\omega_2 - \omega_1)\varepsilon}{((x - a_1)^2 + \varepsilon)^{\frac{1}{2}} \left(((x - a_1)^2 + \varepsilon)^{\frac{1}{2}} + a_1 - x \right)} \right| \\ &\leq \frac{1}{2} \left| \frac{(x - a_1)(\omega_2 - \omega_1)\varepsilon}{((x - a_1)^2 + \varepsilon)^{\frac{1}{2}} ((x - a_1)^2 + \varepsilon)^{\frac{1}{2}}} \right| \\ &\leq \frac{1}{2} \left| \frac{(x - a_1)(\omega_2 - \omega_1)\varepsilon}{(x - a_1)^2} \right| \\ &= \frac{1}{2} \left| \frac{(\omega_2 - \omega_1)\varepsilon}{x - a_1} \right| \end{aligned}$$

2) When $a_{i-1} < x \leq a_i$ ($i \geq 2, i \in \mathbb{N}$),

$$\left| \hat{f}_{\varepsilon_i}(x) - f_i(x) \right| = \left| \frac{(\omega_i + \omega_{i+1})x + \sum_{t=1}^{i-1} 2(\omega_t - \omega_{t+1})a_t + (\omega_i - \omega_{i+1})a_i}{2} + \frac{(x - a_i)^2(\omega_{i+1} - \omega_i)}{2((x - a_i)^2 + \varepsilon)^{\frac{1}{2}}} - \omega_i x - \sum_{t=1}^{i-1} (\omega_t - \omega_{t+1})a_t \right|$$

$$\begin{aligned}
&= \frac{1}{2} \left| \frac{((x-a_i)^2 + \varepsilon)^{\frac{1}{2}} - (x-a_i)}{((x-a_i)^2 + \varepsilon)^{\frac{1}{2}}} (x-a_i)(\omega_{i+1} - \omega_i) \right| \\
&\leq \frac{1}{2} \left| \frac{(x-a_i)(\omega_{i+1} - \omega_i)\varepsilon}{((x-a_i)^2 + \varepsilon)^{\frac{1}{2}}((x-a_i)^2 + \varepsilon)^{\frac{1}{2}}} \right| \\
&\leq \frac{1}{2} \left| \frac{(x-a_i)(\omega_{i+1} - \omega_i)\varepsilon}{(x-a_i)^2} \right| \\
&= \frac{1}{2} \left| \frac{(\omega_{i+1} - \omega_i)\varepsilon}{x-a_i} \right|
\end{aligned}$$

B.1. PDF of initial charging time

$$f_{t_s}(x) = \begin{cases} \frac{1}{\sigma_s \sqrt{2\pi}} e^{-\frac{(x-u_s)^2}{2\sigma_s^2}} & (u_s - 12) \leq x \leq 24 \\ \frac{1}{\sigma_s \sqrt{2\pi}} e^{-\frac{(x+24-u_s)^2}{2\sigma_s^2}} & 0 \leq x \leq (u_s - 12) \end{cases}$$

where $u_s = 17.6$, $\sigma_s = 3.4$.

B.2. PDF of EV mileage and initial charging time

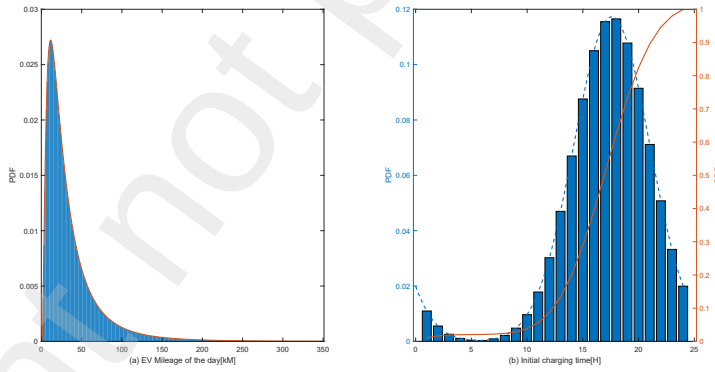


Figure 21 PDF of EV mileage and initial charging time

B.3. Multiple EVs generated by MCS

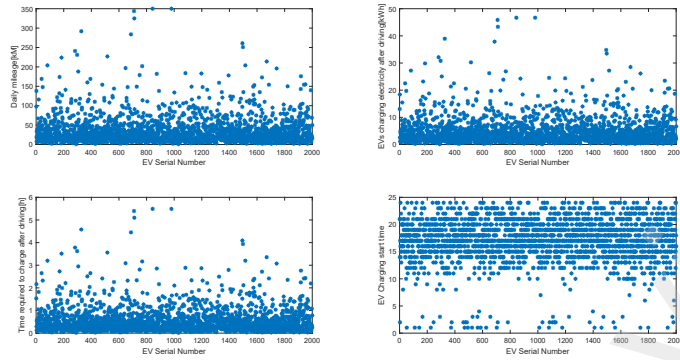


Figure 22 Multiple EVs generated by MCS

B.4. Stochastic charging demand flow diagram

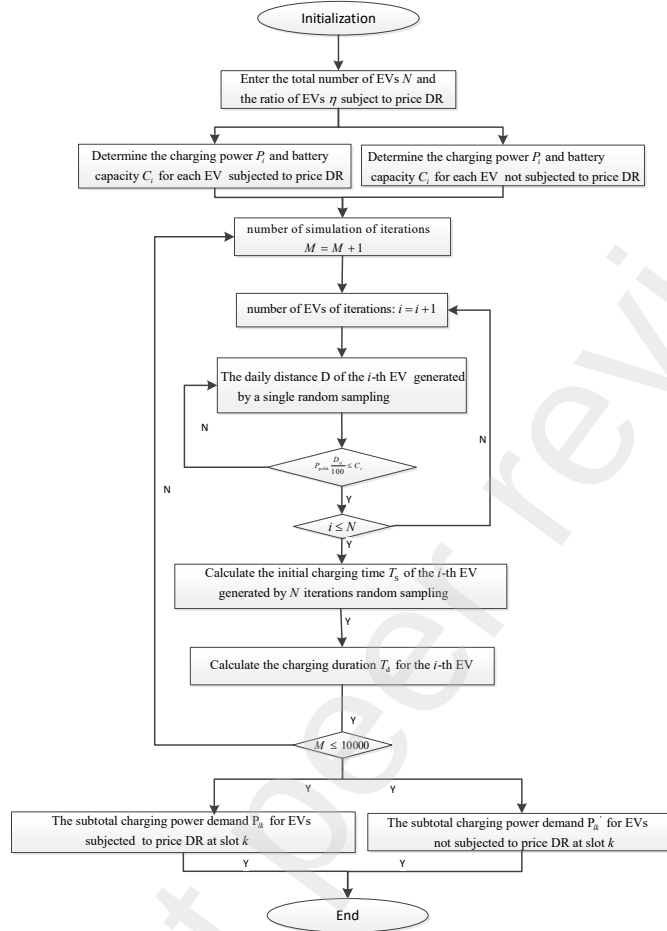


Figure 23 Stochastic charging demand flow diagram for EVs obtained using MCS

B.5. Sensitivity analysis of EV charging load using MCS

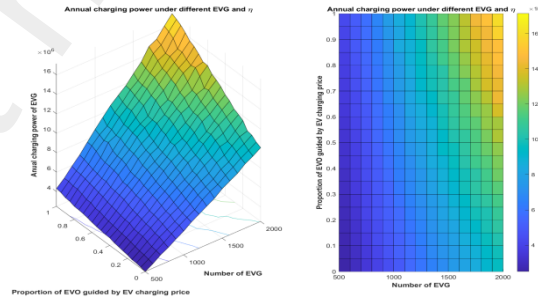


Figure 24 Sensitivity analysis of EV charging load using MCS

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